

```

1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4
5  ENTITY s24_3b IS
6      PORT (
7          clk : IN STD_LOGIC;
8          d_in : IN STD_LOGIC;
9          sel : IN STD_LOGIC;
10         y : OUT STD_LOGIC_VECTOR(4 DOWNTO 0)
11     );
12 END ENTITY;
13
14 ARCHITECTURE behavioral OF s24_3b IS
15     SIGNAL q : STD_LOGIC_VECTOR(4 DOWNTO 0);
16 BEGIN
17
18     -- s_reg process
19     s_reg_p : PROCESS (clk)
20     BEGIN
21         IF rising_edge(clk) THEN
22             IF sel = '1' THEN
23                 q <= q(3 DOWNTO 0) & d_in;
24             ELSE
25                 IF d_in = '0' THEN
26                     q <= q(2 DOWNTO 0) & "00";
27                 ELSE
28                     q <= q(2 DOWNTO 0) & "11";
29                 END IF;
30             END IF;
31         END IF;
32     END PROCESS;
33
34 END ARCHITECTURE behavioral;

```